On Job Training - 1

1.Select any one job role of his/her interest in an organization for next 4 weeks and submit a report to the training supervisor and copy to cohort owner focusing on:

Job Role: AI/ML Intern (Backend Development & Automation)

This internship focused on developing an automated system for processing SMS data using Python, APIs, and Automation Anywhere A360. The project aimed to design a complete data processing pipeline that integrates data extraction, preprocessing, transformation, and report generation into a single automated workflow.

The system developed during the internship, known as the **GP-SMS Automation Pipeline**, enables automatic downloading of SMS data from Google Sheets, cleaning and processing the data, merging it with master datasets, generating structured Excel reports, and sending automated email notifications.

I. Application of Skills and Technical Knowledge

During the internship, a wide range of technical skills and tools were applied to develop the system.

- **Programming Language: Python**

Python was used as the primary programming language for backend development and automation. It was used for writing scripts, processing data, and implementing business logic.

- **Libraries Used: Pandas**

Pandas was used for data processing tasks such as:

- Data cleaning
- Removing duplicates
- Handling missing values
- Data transformation and structuring

- **API Development & REST Concepts**

I developed multiple APIs using REST architecture. These APIs handled:

- Data retrieval
- Data insertion
- Data updating
- Data deletion

APIs developed:

- HRMS API ,Banking API ,Leave Management API ,Warehouse API

➢ Authentication & Security

Implemented:

- Token-based authentication (JWT)
- Session-based authentication
- API key authentication

➢ ASGI Server: Uvicorn

Used Uvicorn ASGI server to run APIs and understand backend execution.

➢ Testing Tool: Postman

Used Postman to:

- Test API endpoints
- Validate responses
- Debug errors

➢ Automation Tool: Automation Anywhere A360

Used TaskBot to:

- Automate complete workflow
- Execute scripts automatically
- Reduce manual intervention

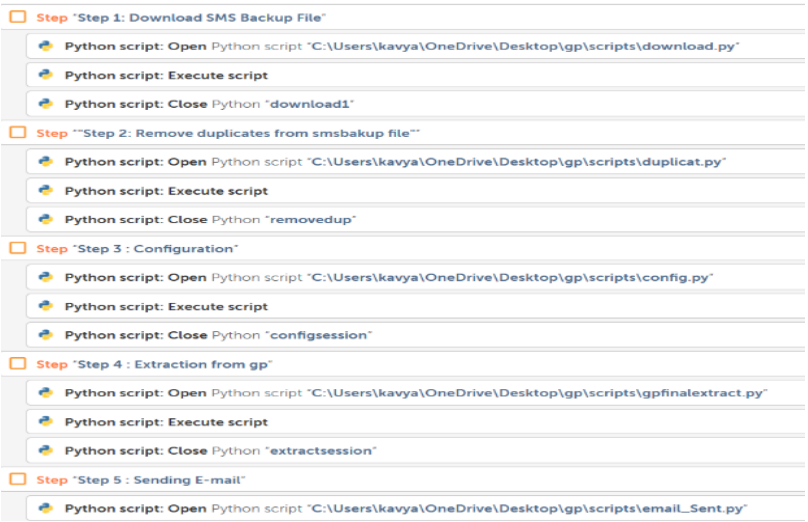
II. Performance on Assigned Tasks and Project

The assigned tasks were completed in a structured and systematic manner. The development followed a step-by-step approach where each component was built, tested, and integrated.The major project handled during the internship was the **GP-SMS Automation Pipeline**.

The workflow included:

- Downloading SMS data from Google Sheets
- Converting file formats
- Cleaning and preprocessing data
- Removing duplicate records
- Filtering new SMS data
- Merging with GP master dataset
- Generating Excel reports
- Sending automated email notifications

The entire workflow was automated using Automation Anywhere A360 TaskBot, enabling execution without manual intervention.



I faced challenges such as data inconsistencies, duplicate records, and API errors. These were resolved through debugging, testing, and iterative improvements.

III. Value Addition to the Organization

The GP-SMS Automation Pipeline transformed a manual data processing system into a fully automated workflow. This reduced time consumption and minimized human errors.

Key contributions include:

- Automated data processing pipeline
- Improved data accuracy and consistency
- Reduced manual effort
- Faster report generation
- Automated email communication

The integration of Automation Anywhere A360 TaskBot enabled complete end-to-end automation, improving productivity and operational efficiency.

IV. Job Role Understanding and Relevance

The role undertaken during the internship was aligned with AI/ML and backend development.

The responsibilities included:

- Data preprocessing and validation
- API development
- Automation implementation
- Testing and debugging
- Report generation

This role helped in understanding how theoretical knowledge is applied in real-world scenarios and how systems are developed in an industry environment.

Use Case - 1

2.1 Use Case Title

GP-SMS Automated Data Processing System

2.2 Description

The system automatically processes SMS data from external sources, performs data cleaning and transformation, merges it with master data, and generates structured reports with minimal human intervention.

2.3 Actors

- System
- User (Admin/Organization)

2.4 Inputs

- SMS data (Google Sheets)

Slot	Address	Name	Date	Type	Body
	+919926075882		2026-03-1	Received	Generally ji itawadi Angio under dr akhilesh jain
	+917772048599		2026-03-1	Received	Premlata gehlot(chomela Mandsaur)Ref to Kokilaben Dhirubhai Ambani Hospital ref by Dr Arjun Gehlot
	+919826550537		2026-03-1	Received	Patient Name-Radheshyam digaRef to kokilaben hospital Dr. Asad Riyaz Ref by Dr Mukesh Jatav
	+917225845233		2026-03-1	Received	Pts name .Gabbu singhRefer to dr.akhilesh jain sirRefer by.dr.deepak jat
	+919977980169		2026-03-1	Received	Patient- Omprakash Soni 66 Male for urology department Referred to Kokilaben hospital Reffering Doctor Wamik Khan
	+918962222700		2026-03-1	Received	Pt arav bariya age 15/m. Ref to kdah indore ref by dr pradeep tiwari
	+919827346544		2026-03-1	Received	PATIENT NAME MRS. POOJA JAIN 30 year REFFER TO KOKILABEN HOSPITAL REFFERING DR DEEPAK JAISWALUNDER DR SUNITA CHOUHAN .
	+919144556441		2026-03-1	Received	Mahendra pandey 62 Nagda for dr Ravi DosiRef to Kokilaben Dhirubhai Ambani Hospital ref by Dr C.k. Mod
	+919425348229		2026-03-1	Received	Pt Angad vora Dr Akhilesh Jain Refer by Dr Manoj Patodi
	+919302040118		2026-03-1	Received	Mr.ANURAG PRATAP NEUROFOR MRIREFERRING DR BHAGWAT SINGH TOMAR
	+919826247391		2026-03-1	Received	Divit panchalRef to kokilaben hospital To ruschira pahareFrom dr Ashutosh parashar
	+919893214459		2026-03-1	Received	Pts name..Nidhi jain.... referrer...Dr akhilesh jain sir... cardiologist.... Refar by..Dr anurag jain Kalani nagar airport road indore
	+919329936080		2026-03-1	Received	Patient Name- Divit panchal Ref to kokilaben hospital Ruchira pahare Ref by Dr. Ashutosh parashar
	+919329936080		2026-03-1	Received	divit panchal for pedo case
	+919926719649		2026-03-1	Received	Mr.Kamal DhanwalAge 60/MRef by Dr.Vinod BhattRef- Kokilaben
	+919926719649		2026-03-1	Received	patient babulal yadav ji refer to kokilaben indore by dr vinod bhatt
	+919926719649		2026-03-1	Received	Pradeep Patidar ji refer to kokilaben indore by dr vinod bhatt
	+919977732567		2026-03-1	Received	Patient name Malti bhai Raghuvanshi ref to- kokilaben Hospital Dr. Pranav Referred by Dr Virendra Raghuvanshi
	+917024460600		2026-03-1	Received	Tomorrow patient will come

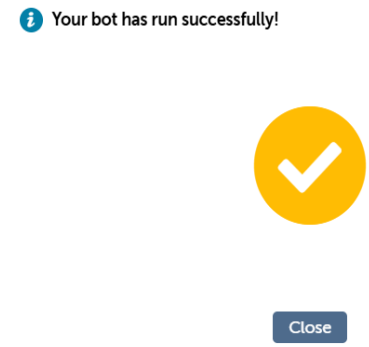
Fig

- GP master dataset
- Date filters
- Phone numbers

2.5 Process

- System downloads SMS data
- Converts file format
- Cleans and preprocesses data
- Removes duplicate records
- Filters new SMS entries
- Merges with GP master dataset
- Generates Excel report
- Executes automation using A360 TaskBot
- Sends email notification

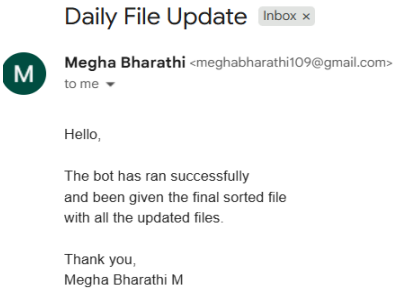
2.6 Output



- Processed Excel report

60496	Nitish Shar	+9191098	New Indor	Suyash Sab	BRAJ MOH	2026-02-0	2026-02-0	11:22	Patient Na	807445	FDHPS472	Nitish Shar	General Pr	NOT_MAPPED
60497	NOT_MAP	CP-CrediH	NOT_MAP	NOT_MAP	NOT_MAP	2026-02-0	2026-02-0	13:54	Patient Na	NOT_MAP	NOT_MAP	NOT_MAP	NOT_MAP	NOT_MAPPED
60498	NOT_MAP	+91998111	NOT_MAP	NOT_MAP	NOT_MAP	2026-02-0	2026-02-0	13:59	Pt Gajendr	NOT_MAP	NOT_MAP	NOT_MAP	NOT_MAP	NOT_MAPPED
60499	Deepanshu	+91947965	Ratlam	Sushil Upa	BRAJ MOH	2026-02-0	2026-02-0	14:52	Mr.Dubey	807443	AQRPT552	Deepanshu	MBBS	NOT_MAPPED
60500	NOT_MAP	+9198262	NOT_MAP	NOT_MAP	NOT_MAP	2026-02-0	2026-02-0	17:03	Refered by	NOT_MAP	NOT_MAP	NOT_MAP	NOT_MAP	NOT_MAPPED
60501	NOT_MAP	+9198262	NOT_MAP	NOT_MAP	NOT_MAP	2026-02-0	2026-02-0	17:03	Pt Noorjah	NOT_MAP	NOT_MAP	NOT_MAP	NOT_MAP	NOT_MAPPED
60502	NOT_MAP	+9189595	NOT_MAP	NOT_MAP	NOT_MAP	2026-02-0	2026-02-0	17:10	Pt name M	NOT_MAP	NOT_MAP	NOT_MAP	NOT_MAP	NOT_MAPPED
60503	NOT_MAP	AD-TRAINI	NOT_MAP	NOT_MAP	NOT_MAP	2026-02-0	2026-02-0	18:34	Please be i	NOT_MAP	NOT_MAP	NOT_MAP	NOT_MAP	NOT_MAPPED
60504	NOT_MAP	+9183599	NOT_MAP	NOT_MAP	NOT_MAP	2026-02-0	2026-02-0	20:38	Patient Jay	NOT_MAP	NOT_MAP	NOT_MAP	NOT_MAP	NOT_MAPPED
60505	NOT_MAP	+9195759	NOT_MAP	NOT_MAP	NOT_MAP	2026-02-0	2026-02-0	09:12	Patient Na	NOT_MAP	NOT_MAP	NOT_MAP	NOT_MAP	NOT_MAPPED
60506	Neha Hard	+9179998	Central Inc	Suyash Sab	BRAJ MOH	2026-02-0	2026-02-0	10:25	Ptnt kanha	807691	AGFPH396	Neha Hard	General Pr	NOT_MAPPED

- Clean and structured data
- Automated email notification



2.7 Advantages

- Reduces manual effort and human errors
- Improves data accuracy and consistency
- Saves time through automation
- Provides real-time reporting
- Enhances productivity and efficiency
- Scalable and reusable system
- Easy integration with other systems

2.8 Limitations

- Depends on data quality
- Requires proper setup
- Limited to structured data

2.9 Future Enhancements

The GP-SMS Automation Pipeline can be further enhanced to improve its functionality, scalability, and efficiency. Several improvements can be implemented to make the system more advanced and user-friendly.

Conclusion

In conclusion, the internship provided valuable experience in applying technical knowledge to real-world projects. The development of the GP-SMS Automation Pipeline demonstrated the effective use of Python, APIs, and automation tools in solving practical problems.